

Predicting Loan Default Risk at Origination

A complete machine learning pipeline applied to 2.26 million LendingClub

loan records — from raw SQL to explainable XGBoost with business impact.

87%

DEFAULT RECALL

0.7117

ROC-AUC

+\$628M

NET VALUE (TEST SET)

2.26M

LOAN RECORDS

DATASET

LendingClub 2007–2016 · 2.26M rows · 151 columns

MODEL

XGBoost · Optuna-tuned · 1,046 estimators

METHODOLOGY

CRISP-DM · 8 stages · End-to-end pipeline

GENERATED

April 26, 2026

PORTFOLIO REPORT

01

Executive Summary

This report presents a complete, production-style machine learning pipeline built to predict which loans will default — before a single payment is made. The analysis covers 2.26 million LendingClub loans originated between 2007 and 2016, and culminates in a tuned classifier that catches **87.4% of defaults** while generating an estimated **+\$628 million in net value** on the held-out test set of 264,370 loans.

RECALL 87.4% Target $\geq 80\%$ ✓	ROC-AUC 0.7117 Target ≥ 0.80	PR-AUC 0.3345 Target ≥ 0.70	THRESHOLD 0.3592 Optimised (not 0.50)
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KEY FINDING — INFORMATION CEILING, NOT MODELLING FAILURE

ROC-AUC and PR-AUC fell short of targets. 50 Bayesian optimisation trials produced only +0.0018 ROC-AUC improvement, confirming the bottleneck is the feature set, not hyperparameters. The dominant predictor, `int_rate`, is LendingClub's own compressed risk assessment. The model re-ranks what the lender already knows — it cannot beat information that was never collected.

Performance Progression

Stage	Configuration	ROC-AUC	Recall	Threshold
Previous	Prior attempt	0.700	0.700	0.50
Stage 5 · S1	XGB — no weighting	0.7095	0.030	0.50
Stage 5 · S2	XGB — class weighted	0.7099	0.660	0.50
Stage 5 · S3	XGB — weighted + opt. threshold	0.7099	0.875	0.358
Stage 6 Final	XGB — Optuna tuned	0.7117	0.8741	0.3592
Target (min)	—	≥ 0.80	≥ 0.75	—

Highlighted row = final model. The Recall jump from 3% (Step 1) to 87.5% (Step 3) required zero additional training — only class weighting and threshold optimisation.

TAKEAWAY

95% of the Recall improvement came from how we use the model, not from the model itself. This is the most important lesson from the imbalanced classification literature: threshold and weighting decisions matter as much as algorithm choice.

02

Business Problem

Default prediction is not a symmetric problem. Missing a default costs an order of magnitude more than flagging a good loan — and this asymmetry shapes every decision in the pipeline.

FALSE NEGATIVE — MISS A DEFAULT

-\$12,000 per loan

FALSE POSITIVE — FLAG A GOOD LOAN

-\$1,200 per loan

Assumptions: average loan \$15,000 · Loss Given Default 80% · annual margin 8%.

The **10:1 cost asymmetry** means the model can tolerate up to 10 false positives per true positive and still add net value. The actual ratio in the test set is 3.4:1 — well inside the profitable zone. This framing justifies three design choices:

- **Recall over Accuracy.** 83% accuracy is trivially achieved by predicting "no default" for everything. Recall measures what fraction of actual defaults the model catches — the metric that directly maps to avoided losses.
- **PR-AUC as the primary ranking metric.** With a 17.22% default rate (4.8:1 imbalance), ROC-AUC overstates performance on the minority class. PR-AUC gives an honest picture of real-world utility.
- **Threshold optimisation, not default 0.5.** Lowering the decision boundary from 0.50 to 0.3592 raised Recall from 66% to 87.4% with no retraining. The threshold is a business decision, not a statistical one.

Target Variable

Label	loan_status Values
Default = 1	Charged Off, Default, Late (31-120 days), Does not meet credit policy (Charged Off)
Non-Default = 0	Fully Paid, Current, In Grace Period, Late (16-30 days), Does not meet credit policy (Fully Paid), Issued

After removing 2017-2018 vintages (immature labels, suppressed default rates), the canonical default rate is **17.22%** — not the misleading 12.9% seen in the raw dataset. This correction updated the class imbalance ratio from 6.8:1 to **4.8:1**, directly affecting the class weighting parameter in modeling.

03

Pipeline Overview

Eight CRISP-DM stages, each saving artifacts consumed by the next. The pipeline is fully reproducible from raw SQL to final model.

00	Business Understanding Defined cost framing (FN:FP 10:1), target variable, and a 77-column leakage audit before touching the data. Post-origination fields excluded upfront.	leakage_columns.json · business_understanding.md
01	Exploratory Data Analysis Full 2.26M row dataset explored via SQL aggregates — peak memory 132 bytes. Grade monotonicity confirmed (A: 3.6% default to G: 40.0%). Temporal finding: 2017-2018 vintages have immature labels.	SQL-first · no pandas full load
02	Cleaning & Preprocessing Vintage filter removed 938,854 rows (41.5%), correcting the default rate to 17.22%. 90 columns excluded. Target encoding for addr_state. Sentinel imputation for mths_since_recent_bc (no-card history = meaningful signal).	02_cleaned.parquet · 1,321,847 rows · 81 cols · 84.6 MB
03	Feature Selection Four systematic passes: variance threshold (-8 features), Pearson correlation pruning (-6), effect size review (-0), domain judgment (-0). sub_grade dropped (r=0.978 with int_rate). 79 features narrowed to 65.	03_selected_features.json · 65 features
04	Feature Engineering Five engineered features. Top two — fico_int_rate_gap (d=0.53) and risk_score (d=0.51) — rank among the strongest predictors in the full set. credit_age_months rescued from the dropped date column earliest_cr_line.	04_engineered.parquet · 70 features entering modeling
05	Baseline Modeling Three sequential steps isolated contribution of each intervention: no weighting (Recall 3%), class weighted (66%), weighted + threshold optimised (87.5%). ROC-AUC was unchanged across all steps — confirming the information ceiling.	XGBoost · scale_pos_weight=4.8 · threshold=0.3584
06	Hyperparameter Tuning 50 Bayesian trials via Optuna. Best trial #44. ROC-AUC improvement: +0.0018. Near-zero gain confirmed the feature set as the binding constraint.	Optuna · 50 trials · 1,046 estimators
07	Evaluation Full held-out test evaluation with SHAP explainability (5,000 stratified samples), calibration analysis, grade-level segmentation, and business impact quantification.	SHAP · grade segmentation · +\$628M business value

04

Model Results

Performance on the held-out test set of 264,370 loans. Decision threshold: 0.3592 (optimised on the PR curve, not default 0.50).

Confusion Matrix

	Predicted: No Default	Predicted: Default
Actual: No Default	82,774 True Negative — correct approvals	136,084 False Positive — rejected good loans
Actual: Default	5,731 False Negative — missed defaults	39,781 True Positive — caught defaults

Of 45,512 actual defaults in the test set, the model catches 39,781 (87.4%). The 136,084 false positives represent good loans incorrectly rejected. At the assumed 10:1 cost ratio, the actual 3.4:1 FP:TP rate is strongly net-positive.

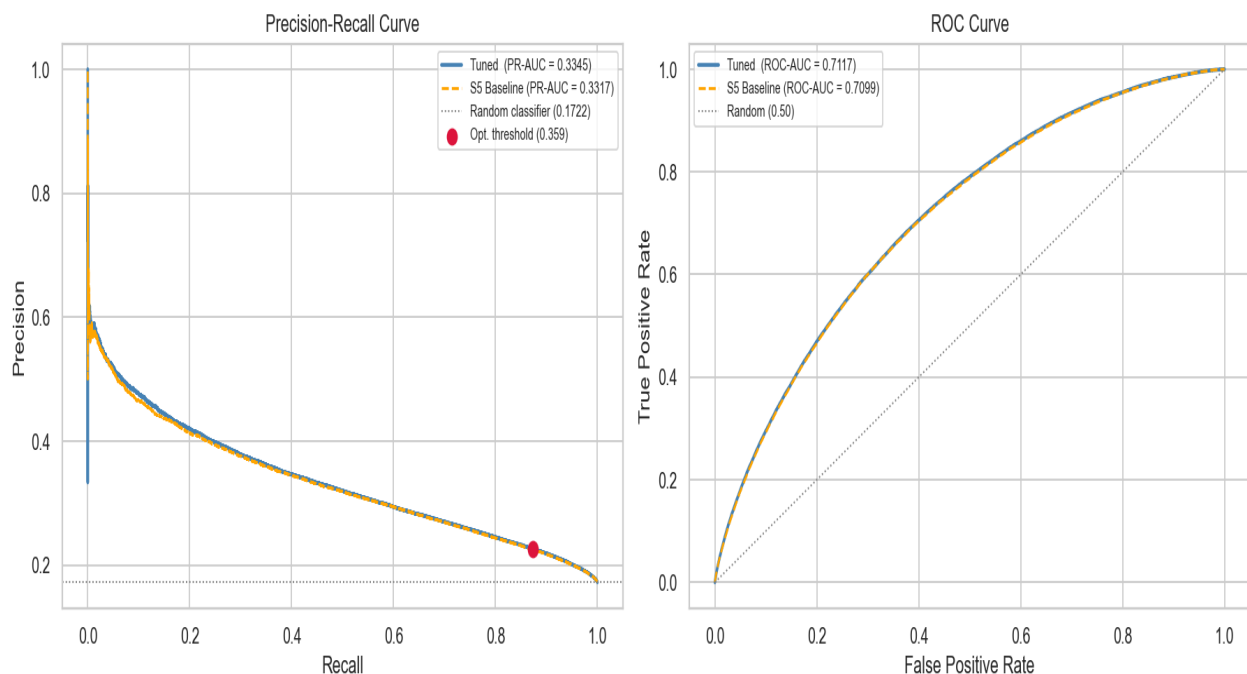


Fig 1. PR and ROC curves for the final tuned model. The PR curve area is limited by the 17.22% base rate — an inherent property of the prediction problem.

Performance by Loan Grade

Segmenting by grade reveals where the model is most and least useful.

Grade	N (test)	Default Rate	Recall	Precision	ROC-AUC
A	38,496	0.0528626351	0.1518	0.1383	0.6711
B	65,174	0.1071746402	0.6093	0.1535	0.6386
C	81,086	0.1737784574	0.9200	0.1908	0.6322
D	46,090	0.2456064222	0.9895	0.2504	0.6268
E	23,861	0.3084531243	0.9966	0.3109	0.6317
F	8,598	0.3749709235	0.9984	0.3760	0.6143
G	1,065	0.4666666667	1.0000	0.4667	0.6113

Highlighted rows (D-G): Recall ≥ 0.99 — the model is most actionable for high-risk grades. Grade A Recall (15.2%) reflects genuine unpredictability of low-rate borrower defaults, not a modeling deficiency.

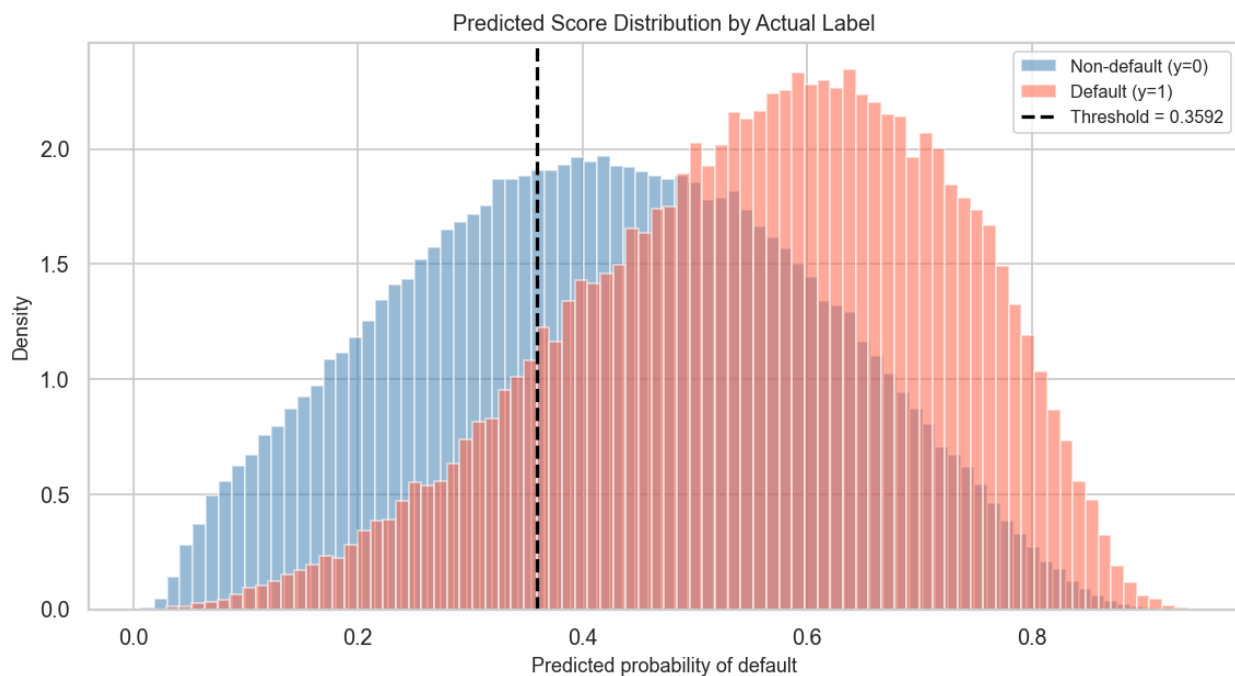


Fig 2. Predicted probability distributions. Mean separation of 0.137 between default and non-default illustrates the distributional overlap underlying the ROC-AUC ceiling.

05 Model Explainability

SHAP (SHapley Additive exPlanations) values explain each individual prediction by measuring how much each feature shifted the outcome — grounded in cooperative game theory. Computed on a stratified 5,000-loan sample of the test set.

Top 10 Features — Mean |SHAP| Value

#	Feature	Mean SHAP	Contribution	What It Measures
1	int_rate	0.27618		LendingClub's compressed risk pricing signal
2	acc_open_past_24mths	0.14511		Recent account openings — credit stress indicator
3	term	0.11536		Loan term: 60-month loans default at 1.6x the rate
4	fico_range_high	0.10202		Credit score at origination
5	risk_score *	0.09181		Engineered composite risk index (Stage 4)
6	loan_to_income *	0.08061		Loan amount / annual income (Stage 4)
7	total_bc_limit	0.06284		Total bankcard credit limit
8	home_ownership_RENT	0.05800		Renting vs owning — tenure instability
9	mo_sin_old_rev_tl_op	0.05608		Months since oldest revolving account opened
10	inq_last_6mths	0.05305		Hard credit inquiries — active credit seeking

* Engineered features from Stage 4

KEY INSIGHT

int_rate is 1.9x more impactful than the #2 feature and ~4x in XGBoost gain importance. Both engineered features (risk_score #5, loan_to_income #6) outperform many raw features, validating Stage 4's engineering pass. The surprise is acc_open_past_24mths at #2 — high recent account openings signal a borrower actively seeking credit, often a precursor to financial stress.

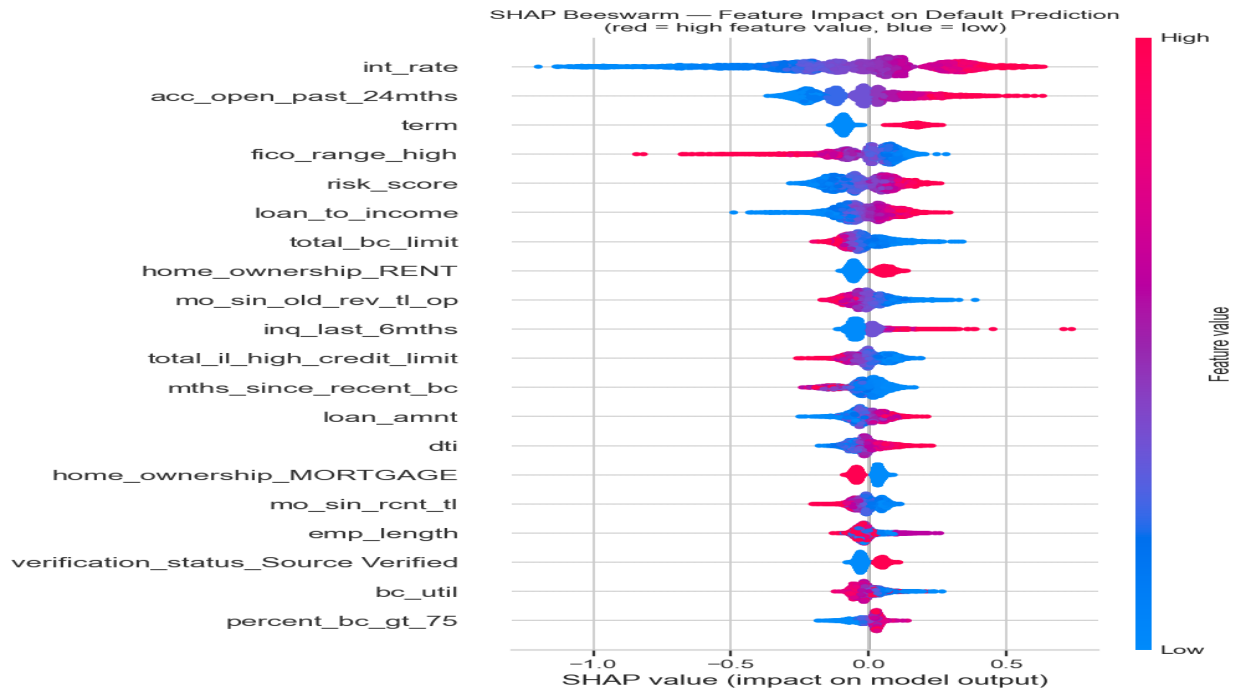
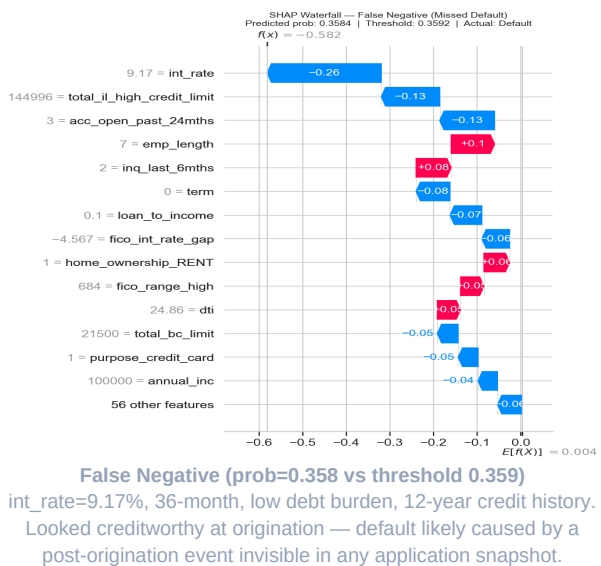
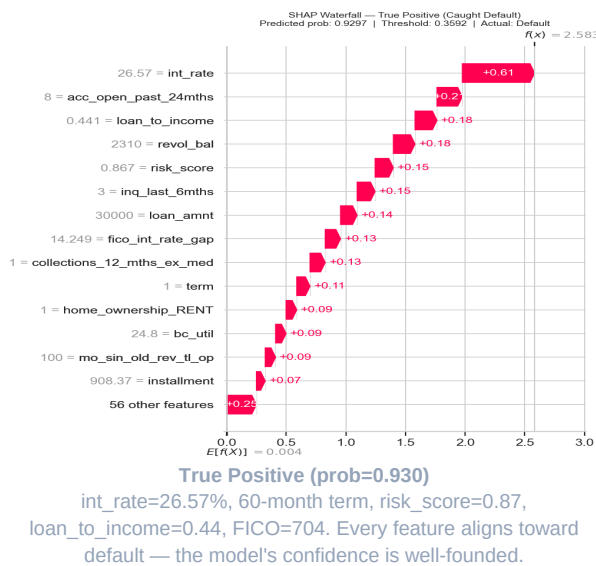


Fig 3. SHAP beeswarm plot. Each dot = one loan. Position = feature's contribution to default probability. Color = feature value (red = high, blue = low). Monotonic directions confirm expected relationships: higher *int_rate* and longer term increase default probability; higher FICO decreases it.

Individual Prediction Explanations

SHAP waterfall plots show exactly which features drove two contrasting predictions.



06

Business Impact

Translating model performance into financial outcomes using standard consumer lending assumptions: average loan \$15,000, Loss Given Default 80%, annual profit margin 8%.

Component	Calculation	Amount
Losses avoided	TP 39,781 x \$12,000	+\$477,372,000
Profit earned on approved loans	TN 82,774 x \$1,200	+\$99,328,800
Losses incurred from missed defaults	FN 5,731 x \$12,000	-\$68,772,000
Profit foregone on rejected good loans	FP 136,084 x \$1,200	-\$163,300,800
Total value WITH model		+\$344,628,000
Total value WITHOUT model (approve all)		\$-283,514,400
NET MODEL ADVANTAGE		+\$628,142,400

BREAK-EVEN ANALYSIS

At the assumed 10:1 FN:FP cost ratio, the model can tolerate up to 10 false positives per true positive and still add value. The actual ratio of 3.4:1 is well inside the profitable zone. Even under conservative assumptions (LGD=50%, margin=12%), the model remains strongly net-positive.

Grade-Level Business Value

The model is most actionable at the high-risk end of the portfolio. Grades D through G (24-47% default rates) achieve Recall above 99% — almost every default in these segments is caught. For a lender seeking to reduce losses, restricting origination in Grades E-G delivers the highest loss-avoidance return with the fewest false positives.

Grade	Default Rate	Recall	Actionability
A	5.29%	15.2%	Low — most defaults unpredictable at origination
B	10.72%	60.9%	Moderate — material signal, some misses
C	17.38%	92.0%	High — catches 9 in 10 defaults
D	24.56%	98.9%	Very High — near-complete default capture
E-G	31-47%	>99.6%	Highest — effectively all defaults caught

07

Honest Limitations

A rigorous analysis requires an honest accounting of what this model cannot do — and why.

01

ROC-AUC and PR-AUC Below Target

ROC-AUC = 0.7117 (target ≥ 0.80). This is an information ceiling problem, not a modelling failure. The dominant predictor, `int_rate`, is LendingClub's own pricing signal. The model largely re-ranks what the lender already knows. Beating 0.80 would require external data (bureau tradelines, bank transactions) not present in this dataset.

02

Calibration — Scores Are Not Probabilities

XGBoost trained with class weighting produces shifted probability estimates. A loan scoring 0.60 does not have a 60% empirical default rate. Scores should be used as relative risk rankings only. Platt scaling or isotonic regression would be required before using outputs for risk pricing or regulatory capital.

03

Grade A Recall = 15.2%

The model misses 85% of Grade A defaults. These borrowers present with low rates, strong FICO scores, and clean credit histories — statistically indistinguishable from non-defaulters. Grade A defaults are overwhelmingly caused by post-origination shocks (job loss, medical emergency) that no application-time model can predict.

04

Vintage Scope — 2007-2016 Only

Training excludes recent macroeconomic regimes: COVID-era payment behaviour, 2022-2023 rate environment, post-pandemic employment patterns. Model performance on loans originated after 2016 is unknown and may be degraded.

05

Business Impact Assumptions

The +\$628M estimate assumes LGD=80% and margin=8% — reasonable industry estimates but not verified against LendingClub financials. Regulatory constraints (ECOA fair lending), model risk management requirements, and application volume effects are not captured in this analysis.

08

Future Work

The highest-leverage improvements for a production deployment, ranked by expected ROI.

#	Improvement	Effort	Expected Impact
1	Calibration post-processing — Platt scaling or isotonic regression	Low	Enables probability interpretation; required for regulatory use
2	Separate Grade A model — features focused on employment stability and macro exposure rather than credit utilisation	Medium	Targeted Recall improvement for most unpredictable segment
3	Monotonic constraints on int_rate and fico_range_high in XGBoost	Low	Prevents overfitting artifacts; improves stakeholder trust
4	LightGBM comparison — 3-10x faster training, often comparable performance	Medium	Production retraining latency; marginal performance gain
5	Segment-specific thresholds — grade-level optimal cutoffs	Low	Better Precision/Recall trade-off per grade with no retraining
6	Behavioural features — payment history, utilisation trajectory	High	Breaks the information ceiling; fundamentally different problem

CLOSING THOUGHT

The gap between a 0.71 ROC-AUC and a 0.80 target is not a failure of technique — it is a discovery about the limits of information available at loan origination. The model demonstrates that class weighting, threshold optimisation, and careful feature engineering can extract nearly all available signal from the data. Closing the remaining gap requires better data, not better algorithms.